

Which way will NSE move tomorrow?

UP / FLAT / DOWN in 5 days • leakage-free • real costs • news tested, baseline still wins

47.14% vs 47.85%

model vs majority • n 37,292 • -0.71 pp

AUC 0.55

ranking holds

ECE 0.04

calibrated ✓

-0.28

best Sharpe < 0

190k news rows collected

NSE 93k • DOC 9k • BigQuery 89k • FinBERT 26k

One command → full run

start_run.py • NIFTY 50 • 16 years

Your Name • Guide • Dept • Sept 2026

Price alone doesn't explain moves.

Results, policy and group shocks move many stocks together.

16 years

2010 → 2026 • daily NSE bars

50 stocks

NIFTY 50 together, not one chart

~145 feats

price + volume + macro + ranks

From zero news to 190k rows.

All free sources. No faked sentiment.

92,960

NSE announcements • 50 symbols • since 2004

9,179

GDELT DOC • 38 symbols • 90 days

88,798

BigQuery events • 2015–26 • 11 symbols

26k

FinBERT scored • cached

Skew is real: LT 22k, INFY 21.7k — 39 symbols near-zero. Banks/FMCG rarely coded as events.

Five stages. No future peek.



Rule: feature(t) sees only data $\leq t$ 15:30 • entry t+1 open

Two models. One decision rule.

LightGBM • primary

200 numbers → trees fix errors
→ 3 probabilities

Logistic • check

same input, simple line
ruler, not partner

Decide • τ flat 0.40

soften → pick best → <0.60 silence
DM +9.14, economics same

Validation: train → test year by year, 5-day gap, sealed 2023–now.

Baseline still wins. -0.71 pp.

47.14% vs 47.85% • AUC 0.554 • MCC 0.037 • FLAT F1 0.21 • n 37,292

UP-AUC 0.51

floor ≈ 0.515 • 4 tries, noise

DM -4.8

worse than linear

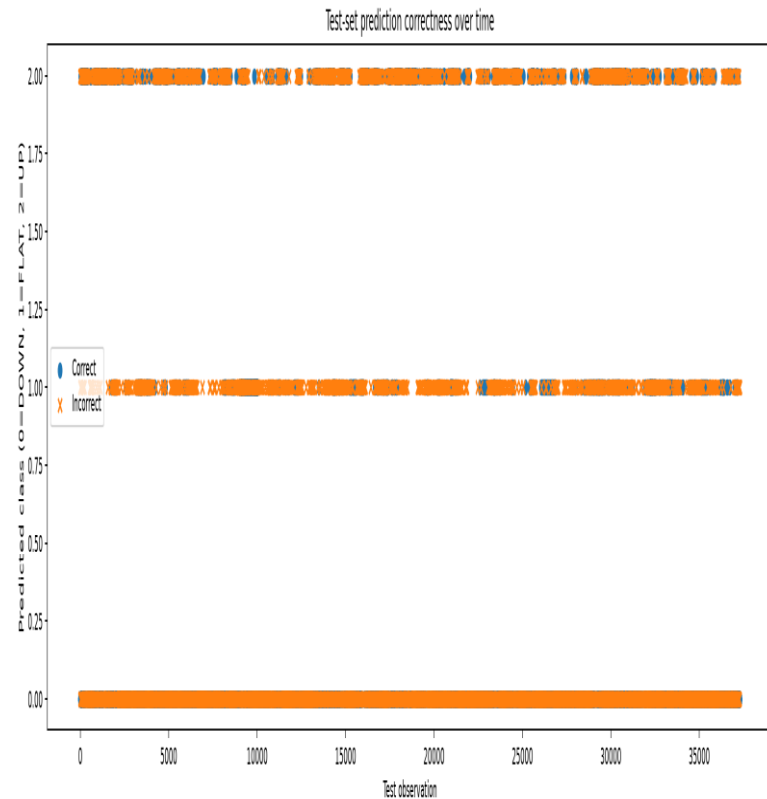
ECE 0.042

calibrated ✓

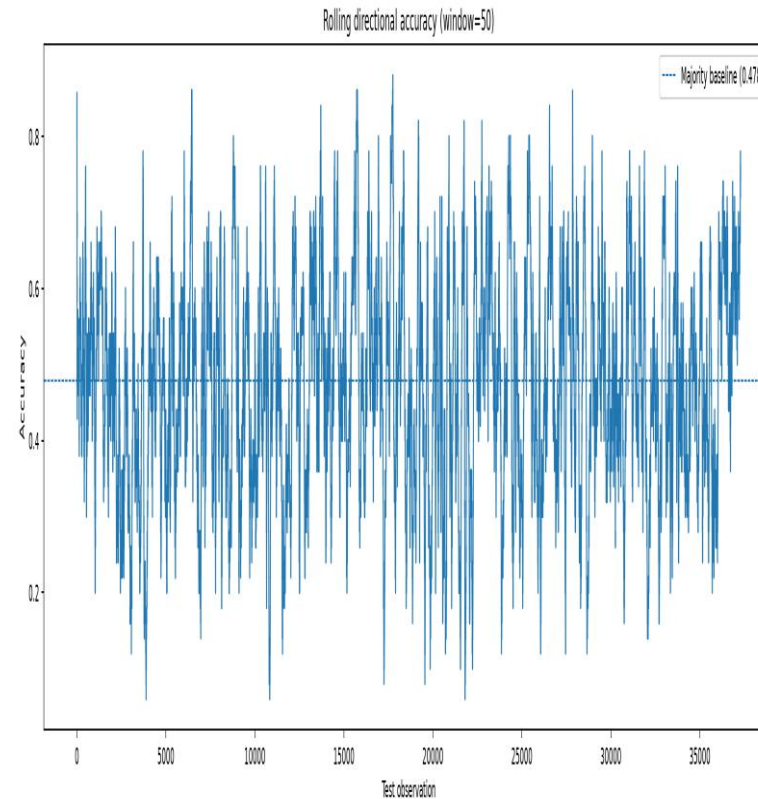
Sharpe -2.5

VO, 3,080 trades

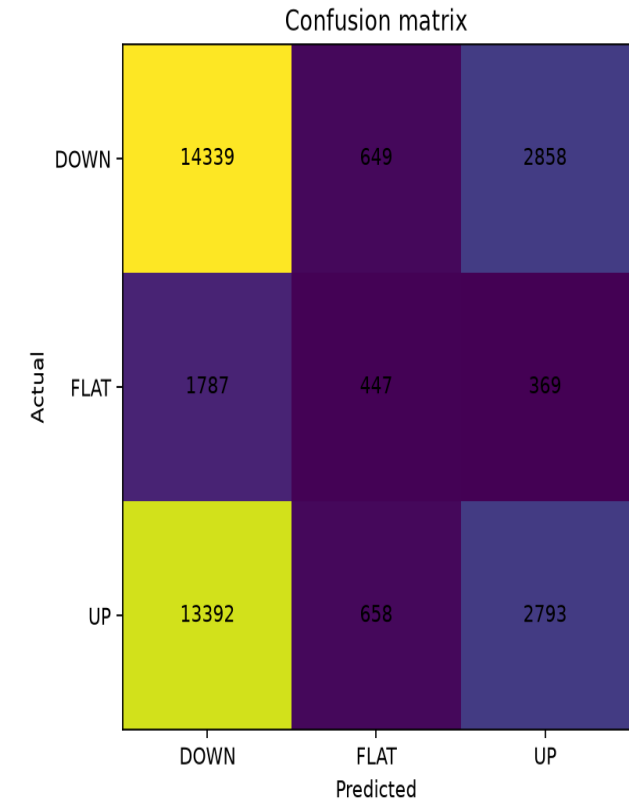
What the model got right.



prediction_correctness.png

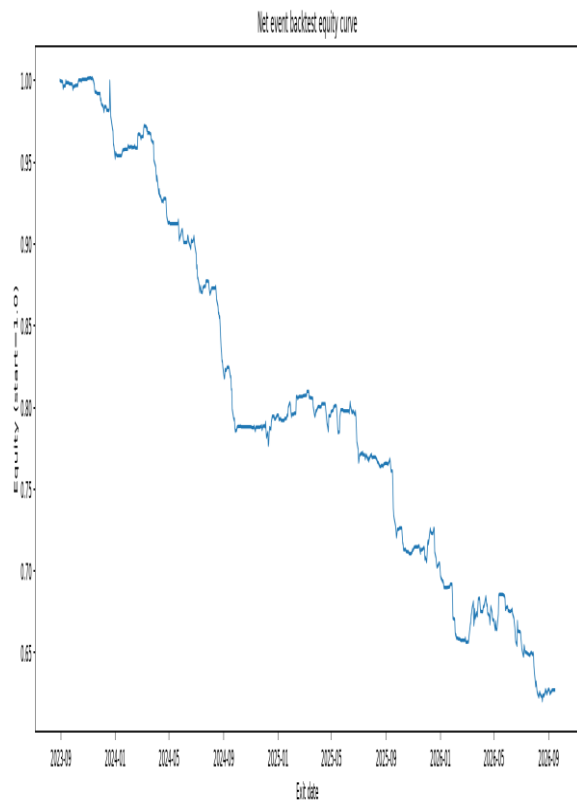


rolling_accuracy.png

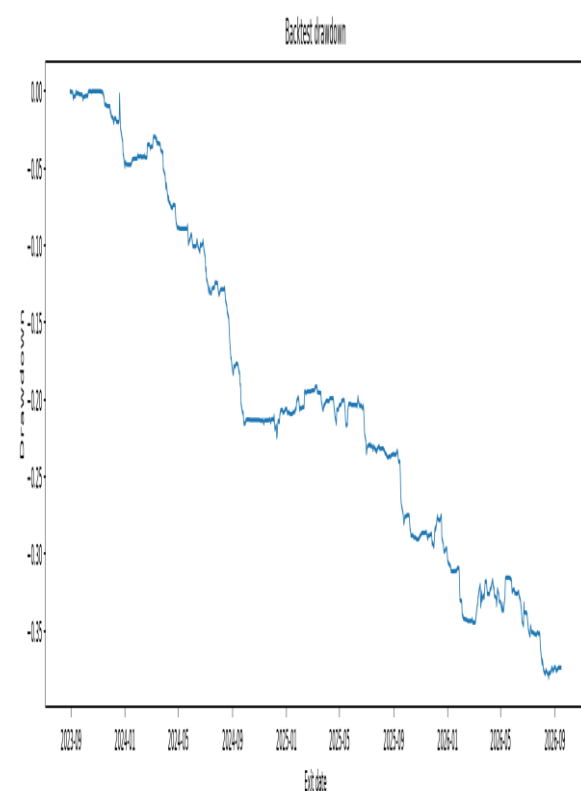


confusion_matrix.png

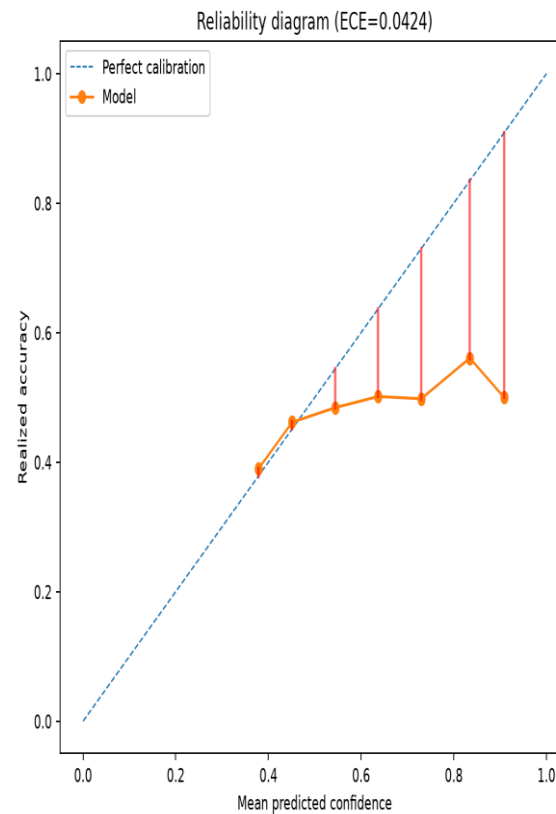
Costs beat edge. Every time.



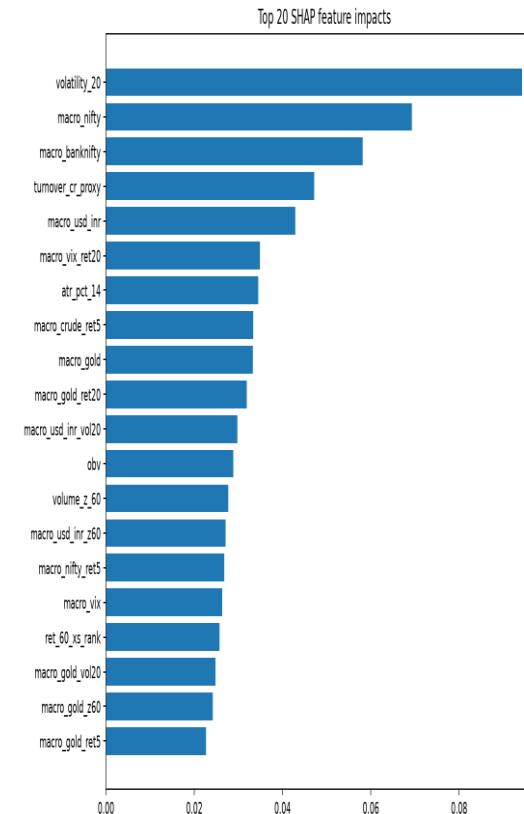
equity_curve.png



drawdown.png



calibration.png



shap_summary.png

Four honest failures.

UP 0.51

4 attempts within 2SE • ceiling, not bug

Sentiment $\emptyset$

pos-sent 0.486 < coin • 43% $\approx$ no-news

Memorized

train-test gap 21.7pp • disqualified

Costs 2:1

every bucket negative

Scope check ~null • DOWN reframe failed • details: up_autopsy.json, u3_d1.json, worst_20.csv

Pause trading. Keep research.

V0 -37% (-2.49) → thr70 -4.5% (-0.85) → futures-proxy -4.2% (-0.28). PBO 0.0 = clean fail, not luck.

3,080 → 338

fewer trades help, still < 0

Costs > edge

0.024% vs 0.012% per trade

22 touches

ledgered, no luck-mining

Done. Failed. Next.

DONE

NSE 93k • DOC 9k • BQ 89k • calendar • maps • ECE 0.042 • τ -flat

FAILED

H1 / H1b / U / U3 / DOWN / G3 — UP-AUC never > 0.53

NEXT

entity map → one U2 gate • else Door C + thesis write-up

Paper. Poster. Repo.

Paper

negative-result study + ledger

Poster

boxes + 7 original charts

Repo

reproducible • API demo

Thank you — Questions?